Jasper Buntinx

Mechanical Engineering · Texas A&M University · B.S. 2027

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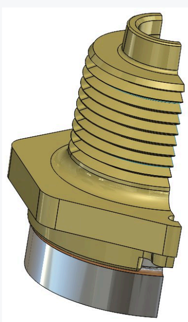
I design hardware and then build it. My work runs from **CAD and engineering analysis** through **CNC programming, design for manufacturability, and fabrication**, into **embedded sensing and controls**, and ends where it should, with a physical machine tested against real requirements. This portfolio covers production engineering at an Amphenol subsidiary, a filament-recycling startup I cofounded, GPU-accelerated CFD research in Brazil, and public-installation steel fabrication.

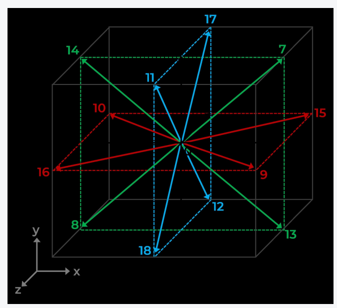
182 → 34 min
CNC CYCLE TIME ON HIGHEST-VOLUME PART

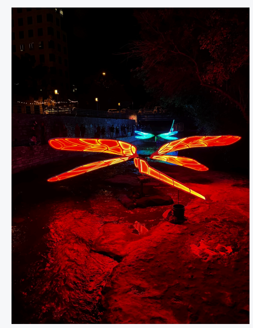
400%+
CFD SOLVER THROUGHPUT GAIN

50-year
FIELD LIFE QUALIFIED VS. 30-YR SPEC

1st place
2024 VEX U STATE CHAMPIONSHIP







CONTENTS

CAPABILITIES

CAD: SolidWorks (CSWA), NX, AutoCAD, OnShape, Fusion 360

CAM: CamWorks, SolidWorks CAM, Fusion 360 CAM, Mastercam

Analysis: GD&T (ASME Y14.5), FEA, ANSYS, CFD, tolerance stack-up

Fabrication: CNC, MIG welding, laser cutting, 3D printing, soldering

Programming: CUDA, C++, Python, Arduino / embedded C

EDUCATION

Texas A&M University · B.S. Mechanical Engineering

College Station, TX · Aug 2023 – 2027 · GPA 3.31 / 4.0

CERTIFICATION

CSWA · Certified SolidWorks Associate

Full case studies and additional media at jasperbuntinx.com

PRODUCTION ENGINEERING · DFM · CAM · QUALIFICATION · AUTOMATION

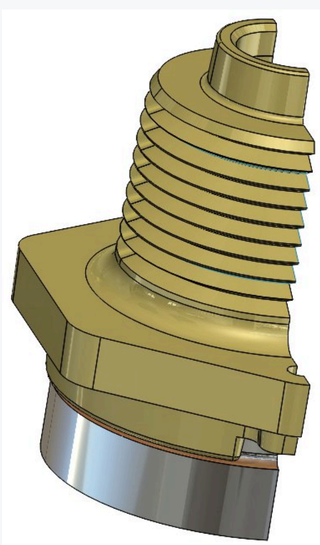
Rochester Sensors (Amphenol subsidiary)

Mechanical Engineering Intern · Dallas, TX · May – Aug 2026

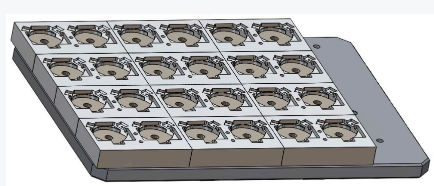
Rochester builds liquid-level and pressure sensing systems for industrial and vehicle applications. I worked across the production lifecycle: developing CNC processes and workholding, troubleshooting in-development designs, qualifying assemblies against their field environment, and automating manual stations.

70–80%	15+	50-year	15–30/min
CYCLE-TIME REDUCTION	PRODUCTION PARTS PROGRAMMED	FIELD LIFE DEMONSTRATED	ROBOTIC BIN-PICK RATE

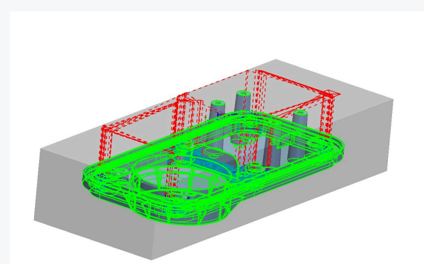
- Programmed CNC machining workflows in CamWorks for **15+ production parts** and dedicated multi-part fixtures, replacing inherited pattern-projected toolpaths to cut cycle times 70–80%, taking the highest-volume part from **182 to 34 minutes**, while holding all GD&T requirements.
- Acted as the **design-for-manufacturability** reviewer between R&D and the shop floor, developing toolpaths and processes for other engineers' prototype parts and housings, and flagging the geometry that quietly costs the most: unnecessary features, extra setups, and small deep pockets that force long-reach cutters to run slow and break mid-job.
- Resolved **tolerance stack-up and assembly fit defects** on in-development products in SolidWorks, validating designs by machining prototypes to released dimensions and correcting a source-design misalignment before mass production.
- Executed environmental qualification testing on sensor assemblies for pre-launch client products: accelerated thermal cycling (**−40 to +125 °C**), salt ingress, and vibration, demonstrating reliability beyond 50-year equivalent field life against a 30-year design requirement.
- Programmed a **Universal Robots arm for vision-guided bin picking** using Keyence 3D scanning, gripping randomly oriented parts from bulk bins and sorting them at 15–30 parts/min, automating the first station of a phased plan to redeploy operator hours.



Machining strategy applied to a production component.



Multi-part workholding designed for a production run.



Finishing passes and stock-removal verification in CamWorks.

TOOLS CamWorks · SolidWorks · GD&T · DFM review · Tolerance analysis · Environmental testing · Universal Robots · Keyence 3D vision

STARTUP · MACHINE DESIGN · THERMAL & CONTROLS

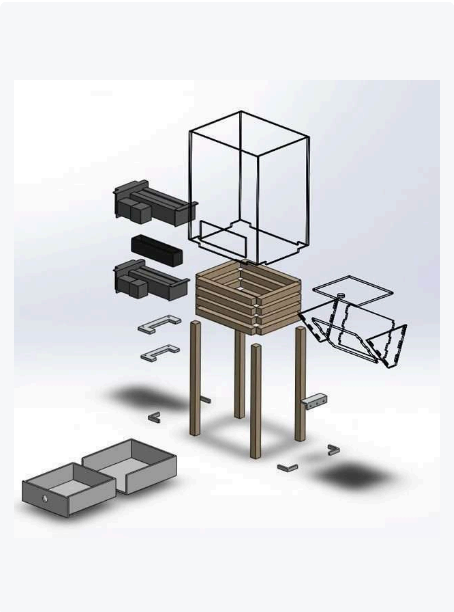
ECOFIL Filament Recycler

Cofounder / Mechanical Engineer · ECOFIL Team, Aggies Create · College Station, TX · Aug 2025 – May 2026

A three-machine line that turns failed 3D prints back into usable filament: a shredder, a dehydrator, and an extruder. I lead mechanical design from CAD and analysis through fabrication, sensing, and system integration. Every machine on this page was designed and built by our team from nothing.

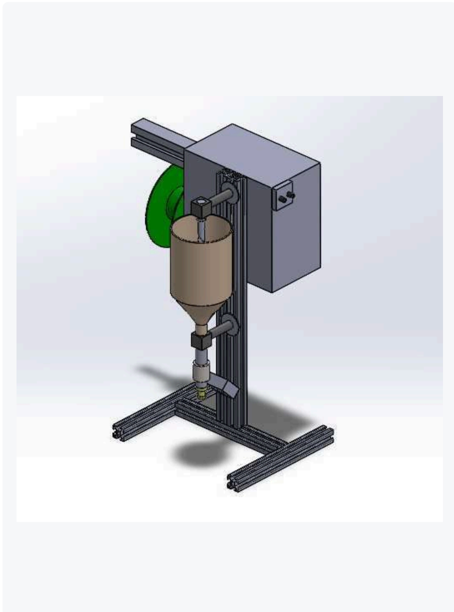
±0.10 mm	3 stages	Closed loop	Funded
EXTRUDED DIAMETER CONTROL	SHREDDER / DEHYDRATOR / EXTRUDER	THERMAL & DIAMETER SENSING	INCUBATOR GRANTS & DONATIONS

- Designed the three-stage recycler in SolidWorks, applying **GD&T to control mating fits** and hold extruded diameter to ±0.10 mm for consistent standard filament quality.
- Fabricated functional prototypes of all three machines through **3D printing, CNC milling, laser cutting, soldering, and MIG welding**, providing working models for performance testing and design iteration.
- Integrated the full control stack myself for real-time monitoring and control of the recycling process: **Arduino controllers, high-voltage AC motors, linear Hall-effect sensors, MOSFETs, PTC heating elements, thermistors, stepper motors, fans, and hygrometers**.
- Secured **incubator funding and material donations** from Aggies Create, Tyrex, and the TAMU Meloy program, then managed project finances and materials through detailed BOMs to complete the recycler within budget.



Shredder assembly, exploded view in SolidWorks.

Dehydrator prototype built from waterjet and welded stock.



Extruder stage with hopper, heater, and spooling.

TOOLS SolidWorks · GD&T · SolidWorks CAM · Arduino / embedded C · MIG welding · Waterjet & laser cutting · BOM management

ENERGY INDUSTRY INTERNSHIP · GPU COMPUTING · FLUID DYNAMICS

Petrobras CFD Optimization

Mechanical Engineering Intern · Petrobras / Geoenergia Lab, UDESC · Florianópolis, Brazil · Jun – Aug 2025

Petrobras needed to predict how spilled oil travels and disperses through seawater, over ocean-scale domains, fast enough to guide a response. The premise was that a GPU-parallelized lattice Boltzmann solver could reach a better speed-to-accuracy point than a traditional Navier-Stokes approach. I profiled and reworked that solver, validating every speedup against the simulation's accuracy limits rather than accepting raw throughput as the result.

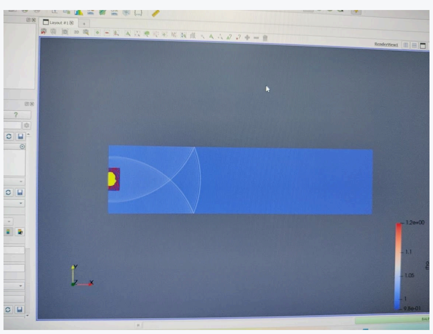
400%+
MLUPS THROUGHPUT GAIN

+60%
WARP EFFICIENCY

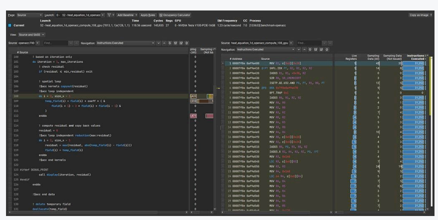
56 → 36 KB
PEAK SHARED MEMORY

+33%
STABLE PEAK REYNOLDS NUMBER

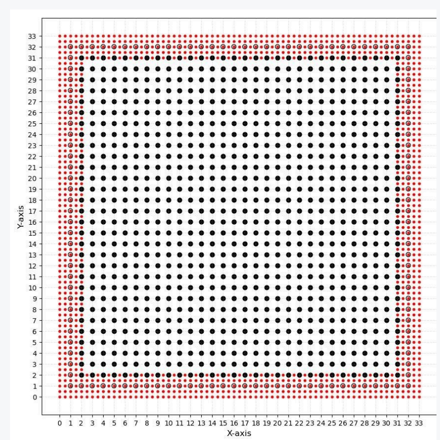
- Optimized CUDA-based Lattice Boltzmann CFD solvers using **AI-assisted kernel profiling in NVIDIA Nsight Compute**, resolving memory bottlenecks to push throughput past published open-source LBM benchmarks.
- Achieved **400%+ throughput gain (MLUPS)** via FP64→FP32 migration and memory-access restructuring, validated against solver tolerances so accuracy was never traded for speed.
- Authored a **Python mesh-generation script** for fine-to-coarse grid interfaces, raising the stable peak Reynolds number 33% with no throughput loss.
- Diagnosed uncoalesced global memory access via Nsight workload reports and remapped thread-to-data access for **60% better warp efficiency**; cut peak shared memory 35% by removing redundant data staging.



Solver output visualized in ParaView.



Nsight Compute kernel profiling used to find bottlenecks.



Fine-to-coarse grid interface from the meshing script.

TOOLS CUDA · C++ · Python · NVIDIA Nsight Compute · Nsight Systems · ParaView · Lattice Boltzmann methods

STRUCTURAL DESIGN · FABRICATION · PUBLIC INSTALLATION

Creek Show Dragonflies

Mechanical Engineering Intern · Waterloo Greenway Conservancy · Austin, TX · May 2022 – Jan 2023

Austin's Creek Show installs large-scale illuminated art over Waller Creek each fall. I translated artist concepts into four illuminated steel sculptures: engineered, fabricated, wired, and installed over a live urban creek, with each dragonfly carrying its own unique wing geometry.

4	14+ ft	200+	−30%
UNIQUE SCULPTURES	WINGSPAN	WELDED JOINTS	PEAK WIND AND FLOOD LOAD

- Derived **four unique SolidWorks assemblies** from Odonata LLC artist samples, enabling production of four large metal dragonfly structures with 14-ft+ wingspans after extensive iteration and prototyping.
- Led fabrication of the full-scale structures: **laser-cut eight unique wing designs** via Fusion 360 CAM, welded 200+ joints, soldered wiring for four LED displays, and produced water-resistant electronics packaging for permanent outdoor exposure.
- Cut **peak wind and flood loads 30%** by swapping hard plastic shells for a microporous ePTFE membrane, then reinforced against peak bending moments with wing reorientation, a widened steel base, and 300 lb of ballast per structure, verified in SolidWorks Simulation.
- Delivered completed installations ready for the show, including on-site assembly and mounting in the creek bed.

The completed installation over the creek at night.

Welded wing frame on the fabrication jig.

On-site installation in the creek bed.

TOOLS SolidWorks · SolidWorks Simulation · Fusion 360 CAM · Fiber laser cutting · MIG welding · LED wiring

PROTOTYPES · PERSONAL BUILDS · COURSEWORK

Additional Projects

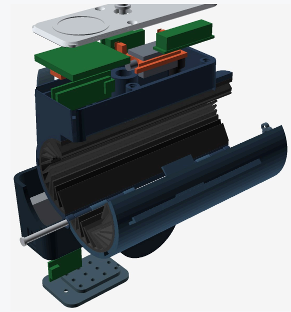
Independent prototypes, competition robotics, and design coursework

RoomCleaner: Laundry-Picking Cable Robot

Personal project · Dallas, TX · May 2026 – Present

Room-scale four-winch cable robot that finds laundry with an overhead camera (YOLO, OpenCV) and carries it to a hamper. Derived the **inverse kinematics**, sized NEMA 17 steppers to 2.6× worst-case tension, and built the fused, kill-switched electrical system with fail-safe homing and a wireless ESP32 end-effector. Modeled 10+ printed parts including a **five-finger tendon-driven TPU gripper**.

SolidWorks · Kinematics · YOLO / OpenCV · Arduino · ESP32 · TPU printing



Automatic Retractable RFID Bike Lock

Personal project · College Station, TX · Dec 2025 – Present

Frame-mounted, NFC-unlocked lock with a **retractable steel cable**: a 15-part 3D-printed clamshell integrating a solenoid latch, PN532 RFID reader, Arduino Nano, Li-ion power, and a spring-loaded spool. PETG housing with a TPU liner fits Ø27–46 mm frames. At its **eighth revision**, with a CNC-milled stainless version next.

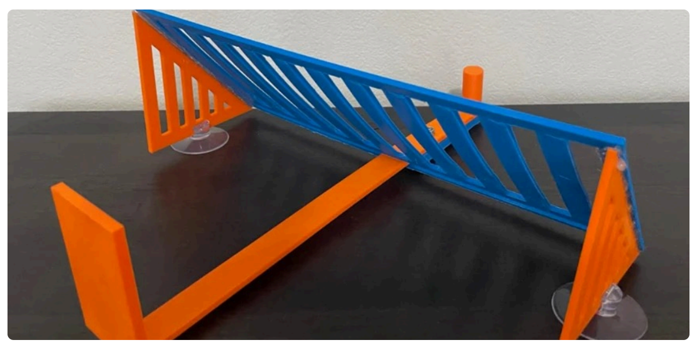
SolidWorks · Arduino Nano · PN532 NFC · Solenoid actuation · PETG / TPU printing · DFM

Gas-to-Electric Tractor Conversion

Personal project · Austin, TX

Full drivetrain swap on a worn-out gas tractor: **tore down the gas drivetrain**, removed the engine and fuel system, and evaluated the transaxle for reuse. Welded in an **electric transaxle and built the battery system and wiring** into the former engine bay, making the machine fully electric.

Drivetrain design · Battery systems · MIG welding · Electrical wiring



Soccer Shots · Tabletop Toy Prototype

MEEN 210 design project · Texas A&M · Jan – Apr 2025

A two-player spring-powered tabletop soccer game **designed for elderly couples and former athletes**, taken from concept sketches to a working all-3D-printed prototype. Built almost entirely in PLA with only springs and suction feet sourced, keeping the build near **$43**.

SolidWorks · User-centered design · DFM · Tolerance fits · 3D printing

COMPETITION & LEADERSHIP

1st Place · 2024 VEX U OVER UNDER State Championship

Aggie Robotics (Team WHOOP5), plus World Championship placement. Contributed AutoCAD schematic redesign and iterative development of an extendable friction launcher to retain, launch, or push Tri-Balls on demand.

Special Events Logistics Chair · One Army Men's Organization

Led large event construction on a $30K+ annual budget, taking builds from design and mockup through BOM and fabrication with 80+ members, supporting Gladiator Dash, a 5K raising $180K+ annually.